

Gesture Ensemble Music System (GEMS)

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Abstract

Gesture Ensemble Music System (GEMS) is a human-machine improvisation environment that integrates three historically significant approaches in musical interaction: ensemble accompaniment technologies, music generation system, and embodied gestural control. Play-along accompaniment systems provide harmonic support, but operate as static recordings or loop-based playback, offering no meaningful adaptation to a performer's phrasing or expressive cues. Modern generative systems can create rich and stylistically coherent output, but are not designed for real-time responsiveness, which is not suitable for interactive improvisation. To bridge these limitations, GEMS uses the concept of gesture control to let the performer's movement directly shape how the generative rhythm section behaves, ultimately achieving both musical flexibility and real-time ensemble interaction.

Keywords

human-machine improvisation, embodiment, gesture tracking, generative accompaniment, interactive music, virtual ensemble, Max/MSP

1 Introduction

In improvised music that centers around harmonic progressions, ensemble members shape form and interaction through a shared rhythmic and harmonic framework. A typical rhythm section, including piano, bass, and drums, supports improvisation by forming the harmonic, rhythmic, and melodic structure of a song. Soloists rely on this communicative process with the rhythm section to improvise and structure their musical phrasing. Any digital system that aims to emulate this environment should therefore account for the adaptive and conversational behavior that characterizes real ensemble performance.

Play-along recordings and digital accompaniment tools have long attempted to simulate this environment for individual practice, offering access to piano-bass-drum backing tracks when collaborators are not available. Yet these systems typically position the improviser as a passive participant: the band plays, and the musician plays over it. The interactive feedback loop that defines real ensemble performance is absent. Despite advances in computer-generated music, this challenge remains unresolved. Although corpus-based and deep-learning systems can generate stylistically convincing music, they are optimized for full-band autonomous generation and operate independently of the performer once the music begins. These models generate music does not support interaction itself: they cannot communicate with the performer for real-time musical change.

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Gesture Ensemble Music System (GEMS) seeks to address this limitation by restoring the interactive relationship between soloist and rhythm section based on the concept of gestural control. Instead of treating accompaniment as a static backing track, GEMS uses gestural input to shape how the virtual ensemble performs in real time. Movement controls musical materials, enabling the performer to influence ensemble behavior as they would with cues in a live setting. Thus, this project explores a system that lets the improviser conduct and converse with a responsive accompaniment system.

2 Background

This section reviews the research landscapes that inform the design of GEMS. We first examine play-along accompaniment systems and modern music-generation technologies, both of which provide useful musical resources but do not yet support fully interactive ensemble performance. We then discuss gesture-controlled musical interfaces, highlighting how embodied control is a possible solution to real-time influence and creative dialogue in performance contexts. By situating GEMS within these three areas of research, we illustrate how it builds on prior work while addressing the specific challenge of achieving responsive, improvisation-ready accompaniment in human-machine collaboration.

2.1 Ensemble Accompaniment

Play-along tools such as the Jamey Aebersold recordings and commercial apps like iReal Pro have long served as valuable practice resources for improvisers. They provide access to a simulated rhythm section when collaborators are unavailable, allowing musicians to explore repertoire and develop fluency over chord progressions. However, their interactivity is fundamentally limited: while users may adjust tempo, key, or style, they cannot influence how each accompanying instrument behaves in response to their musical ideas. As Johnson [7] notes, recent AI-enhanced play-along systems produce more lifelike textures than static recordings, yet still position the musician as a soloist playing alongside an automated track rather than engaging in a collaborative exchange. Earlier jazz education technologies likewise emphasized aligning to a pre-determined accompaniment [8], preserving access to ensemble sound but not the adaptive communication that defines live improvisation.

2.2 Generative Systems

The development of computer music generation reflects a transition from systems grounded in music-theoretic rules to those driven by large-scale data modeling. Early rule-based systems such as Hiller and Isaacson's Illiac Suite explored harmonic logic and contrapuntal structure [18]. Cope's EMI system continued this approach by abstracting and recombining stylistic patterns from symbolic corpora [4]. Lewis's Voyager expanded these principles into real-time contexts, envisioning the computer as an improvising partner with its own musical agency [10].

Recent corpus-informed improvisation tools, such as Somax, preserve real-time operation but remain bound to their datasets, reorganizing pre-existing content rather than generating new harmonic direction [2]. This trajectory intensifies with deep learning models, which prioritize statistical prediction over explicit reasoning about form or ensemble roles [3, 6]. Systems like MusicGen can synthesize convincing stylistic texture, but their responses are not easily or immediately alterable based on performer intent [5].

As Cádiz et al. note, contemporary generative systems interpolate from prior examples rather than interpret expressive cues or structural meaning in the moment. They can produce musically rich output, yet lack mechanisms for rapid adaptation, which is a key requirement in improvisational settings where ensemble members continuously respond to each other’s actions.

2.3 Embodied Interaction and Gestural Control

Gestural interaction in music foregrounds the role of bodily movement in expressing musical meaning [9]. Gesture characteristics are defined mapping movement to sound synthesis parameters using sensors and computer interfaces [15]. These systems increase visibility, clarity, and performer agency in electronic music by situating the body as a primary control source [19].

As Wanderley notes, gestural interaction in musical systems has historically developed along many complementary trajectories [16]. The first involves instrument-like gestural controllers in which performer movement directly produces or modulates sound. Systems in this category emphasize tangible engagement and immediate audiovisual feedback [14], though some intentionally obscure gesture–sound causality for dramatic effect [13].

The second trajectory explores device manipulation for higher-level musical control, where gesture shapes structural behaviors rather than individual sonic events. Early systems such as the Radio Drum [1] and the Digital Baton [11] drew upon conductor-ensemble metaphors, enabling gesture to indicate timing, articulation, or sectional transitions within a predefined musical sequence. Tools like Mathews and Moore’s GROOVE system [12] and Wessel and Wright’s “dipping” metaphor [17] further established gesture as a controller of phrase-level and ensemble-level behavior. In the dipping analogy, the performer’s gesture enters and withdraws from influence, like a hand dipping into water, to continuously adjust the higher musical structure decisions rather than low level musical features like pitch and production parameters.

GEMS is inspired by this lineage of ensemble-level gestural control but extends it into real-time improvisational contexts. Rather than playing back a fixed sequence or score, gestures in GEMS act upon a conductor's hand, direct the harmony and rhythm as the performance unfolds. The introduction of probabilistic variation allows gesture to influence not only structural activation but also the variation of the instrument that characterize live ensemble improvisation. By enabling performers to guide musical decision-making, GEMS positions gesture as a communicative channel for interactive machine musicianship, therefore bridging the gap between static accompaniment and slow-reacting generative systems.

3 System Design

GEMS is implemented in Max/MSP and functions as a real-time generative accompaniment system controlled through performer

movement. The system contains three generative player engines (Chord, Bass, and Drums) that produce MIDI note sequences in response to a user-defined chord progression. Chord input is entered as text and mapped to MIDI note stacks representing voicings with specific chord qualities and inversions. These notes are then scheduled by player engines using default rhythmic templates, which determine when each note or chord should be played within the bar. All engines support two modes of behavior: a stable preset pattern that defines the core groove, and one-bar variations triggered via gesture-based control. The performer's body is tracked in real time using MediaPipe, and upper-body motions activate or mute engines and introduce generative fills that automatically reset to the preset at the next bar boundary. Finally, MIDI outputs from all engines are routed into Ableton Live for sound synthesis and playback, giving performers full flexibility over instrument timbre during performance.

3.1 Player Engine

3.1.1 Chord Engine. The chord engine interprets the symbolic chord progression entered by the user and converts each chord label into a stack of MIDI notes that represent an appropriate voicing. This mapping accounts for chord quality, inversion, and harmonic extensions. The engine follows the global progression in real time and updates its voicing output whenever a new chord event occurs, ensuring that the accompaniment remains harmonically aligned to the user's sequence. Once translated into MIDI, the voicing is played back according to a rhythmic template, which determines when each chord is articulated within the bar. Users may edit the rhythmic placement, and gesture-triggered fills can introduce temporary rhythmic variations.

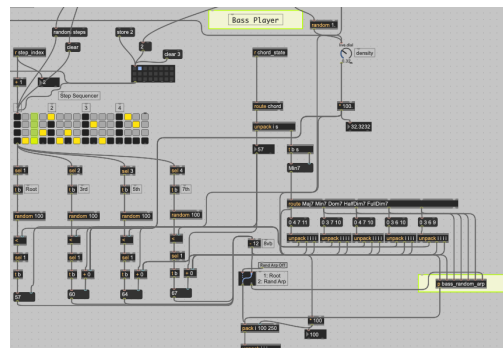


Figure 1: Bass engine player in Max

3.1.2 Bass Engine. The bass engine derives its musical material from the same symbolic chord progression as the chord engine, but reduces the harmony to a monophonic representation suitable for bassline generation. From each chord symbol, the engine identifies key chord tones, such as the root, third, fifth, and seventh, and maps them into a low-register MIDI range appropriate for bass performance. These tones are then sequenced according to a rhythmic template that defines the groove and note placement within each bar. Figure 1 shows the bass engine logic in Max. Like the chord engine, the bass engine responds in real time to changes in the global harmonic progression, updating its pitch material as harmonic events shift. Users may directly edit or fully replace the template with their own programmed bassline patterns. Gesture-triggered commands can introduce temporary probabilistic variations, which remain active only for

the duration of the current bar before the pattern resets to the saved user or default version.

3.1.3 Drum Engine. The drum engine operates independently from harmonic information and instead relies solely on the rhythm template to structure its output. Users can either program their own drum patterns in the sequencer or use a default groove provided by the system. Each bar contains a stable core pattern consisting of kick, snare, and hi-hat, which maintains a consistent rhythmic foundation throughout performance. Additional percussive elements are layered on top using probabilistic triggers that generate short, spontaneous fills. These variations resets automatically at the beginning of each new bar to preserve the groove.

3.2 Gestural Control

Gesture control in GEMS is implemented using a Python module that processes the performer's movement in real time through Google's MediaPipe library. MediaPipe provides a lightweight and efficient computer-vision framework for pose estimation, capable of tracking thirty-two skeletal landmark points across the upper and lower body using only a standard webcam. These tracked points include the head, body, and all major joints, making it possible to capture expressive arm and hand motions without requiring any wearable sensors. Figure 2 shows an example of gestural tracking with a performer playing. For GEMS, three key landmarks are used: the performer's nose, right wrist, and left wrist. Their positional data is streamed from Python into Max/MSP via a UDP receiver.

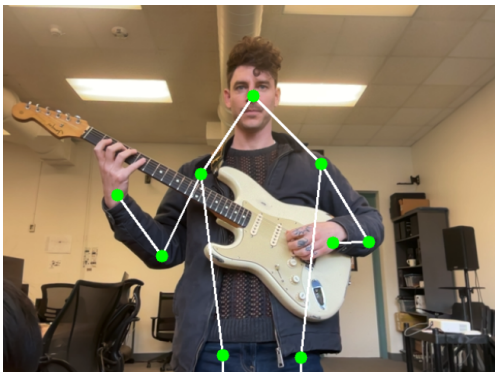


Figure 2: Gesturing tracking using MediaPipe landmarks

3.3 Gesture-to-Music Mapping

Performer movement directly controls the behavior of the generative engines through a set of spatially mapped gestures. Each gesture is defined using the tracked positions of key upper-body landmarks, and each gesture results in an immediate musical response. The performer's nose serves as the master switch for the system: a downward, bow-like motion toggles overall play-back on or off. This gesture mimics a conductor's cue, signaling the beginning or end of a performance segment. The right wrist controls the bass engine. When the wrist moves into the upper-right region of the camera frame, the bass engine is activated or muted. When the wrist moves toward the lower-right region, the system introduces a temporary variation in the bass template: a probabilistic process determines whether each scheduled note is played, while also allowing pitch-level randomness among the

available chord tones. This creates subtle rhythmic and melodic deviations that reset to the preset bassline at the end of each bar. The left wrist controls the chord engine in a similar way. Moving the wrist into the upper-left region activates or deactivates the chord comping pattern. Moving it into the lower-left region triggers a one-bar rhythmic variation where chord attacks follow a probability-driven pattern, creating dynamic shifts in the groove before reverting back to the default rhythmic template on the next downbeat.

By integrating these components, GEMS aligns the three modalities outlined in the related work. Instead of mapping gestures directly to sound parameters or pitch events, GEMS uses body movement to control the underlying generative processes, determining how the ensemble produces music in real time. Through its harmonic pattern generation and probabilistic variation, it builds on generative systems that prioritize musical structure and real-time responsiveness over data imitation alone. And by enabling performers to influence accompaniment roles dynamically, it transforms play-along practice into a collaborative, dialogic exchange more akin to live ensemble improvisation.

4 Evaluation

The initial prototype of GEMS has undergone iterative testing through informal performance sessions with the authors. Both authors are trained musicians, one on guitar, the other on tenor saxophone, with proficiency in improvisation. These instruments require distinct physical approaches to gesture-based control: guitar performance typically involves minimal upper-body movement while maintaining hand position on the instrument, whereas saxophone playing incorporates more pronounced postural shifts and breathing-related motion. This difference in performative embodiment provided contrasting contexts for evaluating gestural mapping strategies. Early evaluations focus on system responsiveness, gestural mapping effectiveness, and the coherence of generated musical output within improvised contexts.

4.1 Technical Performance

The system demonstrates low latency through its MediaPipe integration, with landmark detection responding to performer movement. In its current state, a common issue is the stability of gesture triggering, as it produces false readings more frequently than desired. Further calibration is required to maintain consistent readings of gesture within the vision tracking frame.

The authors used a standard 1080p web camera suitable for initial development stages. Improved camera specifications in future iterations will potentially address current tracking inaccuracies. Additionally, the physical landmarks currently tracked (left arm, right arm, and face) are deliberately limited in number to reduce cognitive load during performance. This design choice is particularly important when considering different instrumental contexts, as guitar, saxophone, and other instruments present distinct gestural affordances that may require instrument-specific mapping strategies.

4.2 Musical Coherence

The system maintains harmonic and rhythmic coherence through its rule-based generative architecture. Because harmonic progressions are determined by user input via the chord progression module, the virtual ensemble consistently follows the intended harmonic structure. This constraint-based approach ensures performance cohesion between the system and human performer,

preventing harmonically unrelated material. Each player engine fulfills a distinct ensemble role: chord engine for harmonic comping, bass engine for chord root and tones, and drums for rhythmic foundation and groove. However, this deterministic approach also limits spontaneous harmonic exploration, as the system cannot deviate from the prescribed chord progression to introduce substitutions or reharmonizations that might emerge in human ensemble interaction.

4.3 Performer Experience

Both authors found the system intuitive to perform with and noted that it maintained a natural ensemble feel comparable to live interaction. Compared to static accompaniment platforms such as iReal Pro or YouTube backing tracks, GEMS provided a more engaging and responsive experience. The probabilistic note selection within each player engine contributed to this sense of liveness, creating variation that reflected authentic ensemble improvisation dynamics. The virtual instruments' timbral characteristics further enhanced this perception, evoking distinct personalities within the rhythm section.

However, gestural control presents a physical constraint: parameter adjustments require hand movements that temporarily reduce instrumental contact. Both guitar and saxophone remain playable with one hand, allowing continued performance during gestural input, though this configuration limits technical facility and affects improvisational flow. This trade-off between gestural expressivity and instrumental technique represents a central design challenge for future iterations.

5 Conclusion and Future Work

GEMS presents an interactive, gesture-controlled performance environment in which distinct player engines form a virtual ensemble. Integrated with DAW-based MIDI instruments, the system enables flexible timbral choices and distinct ensemble member personalities. Building on a long history of gesture-to-music mapping, GEMS extends these principles into the realm of real-time machine improvisation. By combining rule-based generative processes with gestural input, the system supports responsive, structurally grounded music generation that more closely mirrors live ensemble interplay than static backing tracks or black-box generative models. In doing so, GEMS contributes a conductor-like paradigm in which performers shape density, phrasing, and instrumental roles through movement, reframing play-along practice as a site of embodied negotiation between human and machine.

Future development will expand the gesture vocabulary beyond the current limited landmark set, enabling instrument-specific mappings that better accommodate diverse embodied performance practices. Improving the robustness and granularity of vision-based control will be essential for reducing false triggers. Through refined thresholds, calibration routines, and potentially higher-quality camera hardware, the system can achieve a more fluid, instrument-compatible gestural feel. An additional soloist module will be introduced as an improvising counterpart capable of functioning as a lead role within the virtual ensemble. To deepen integration with production workflows, the system will be reconfigured as Max for Live devices, as either a single full-system device or as separate devices for individual player engines.

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